

Basic Automatic Transfer Switch Functions

Technical Bulletin

Publication TB24003 R1



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Certain aspects of transfer switch design attract particular attention, resulting in comparisons of modes of operation ([Automatic](#), [Non-Automatic](#), [Manual](#)), transition sequences ([Open](#), [Delayed](#), [Closed](#)), and special equipment arrangements ([Service Entrance](#), [Bypass-Isolation](#), [Custom Engineered](#)). Instead, this document focuses on the most essential functions that all [Automatic Transfer Switches](#) (ATS) must complete to transfer loads without human intervention.

Essential ATS Function

Regardless of design and feature set, there are specific tasks that an ATS must perform to execute a switching cycle. The tasks required to switch load from a normal power source to an emergency source and back again are explained in the following sections.

Functions and Their Importance

Function 1: Carry Current Continuously

The primary reason why a facility installs a transfer switch is to keep its electrical loads connected to either of two sources of power. Consequently, a transfer switch is designed to remain connected to power distribution circuits under all foreseeable circumstances, including very large fault and short-circuit currents. Conversely, a transfer switch is not designed to open to protect load circuits and equipment. Rather, it is the explicit function of overcurrent protection devices to open circuits, clear faults, and protect downstream circuits and equipment.

Consider the consequences if transfer switch contacts opened before overcurrent protection devices opened as the result of a fault current. First, the switch would be isolated from all sources of power immediately following the event. This contradicts the reason for installing an ATS – to maintain power flow to loads. Second, where overcurrent protection devices are selectively coordinated across a power distribution system, the opening of transfer switch contacts could result in power loss to an unnecessary quantity of loads, increasing the impact on facility operations and making recovery more complex. For more information, see the ASCO White Paper entitled [Selective Coordination Basics](#).

For these reasons, an ATS must be able to continuously carry both 100 percent of its rated service current as well as any fault currents that could reach the transfer switch location. In North America and regions that follow North American standards, Withstand and Closing Ratings that indicate the fault current capacity are verified by testing transfer switches to UL 1008 requirements. For more information, see the ASCO White Paper entitled [UL1008 Transfer Switch Withstand and Closing Ratings](#). For Europe and other regions, see the IEC 60947-6-1 standard or ASCO's [Selecting IEC 60947-6-1 Transfer Switches for Motor Loads in Mission-Critical Applications](#).



Function 2: Detect Power Failures

The capability that differentiates an ATS from non-automatic and manual transfer switches is the ability to self-initiate load transfer when unacceptable power conditions occur on the primary power source. To do so, an ATS must monitor voltage and frequency on the connected source. When voltage and frequency are out-of-range or absent, a transfer switch will initiate a transfer sequence that will connect the load to a secondary source.

Function 3: Initiate the Alternate Source

In a small minority of installations, an ATS may be used to switch between two sources of power that are expected to always be live, such as between two utility feeds. However, for the vast majority of backup power systems, emergency power will be provided by a genset that is driven by an internal combustion engine that operates only when needed. Consequently, an engine(s) must be successfully started so that load can be transferred to the alternator(s) it powers. To do so, an ATS must issue a start signal to the generator set.

Function 4: Transfer Load to the Emergency Source

A genset takes time to start and then accelerate to the operating speed needed to produce acceptable power. The ATS monitors the voltage and frequency coming from the backup generator to verify acceptable power, then transfers load. If the ATS does not detect acceptable power, it will not transfer because this action would not improve power availability to loads. Doing so would leave loads isolated from the normal source should acceptable power return on the primary power source.

The transfer sequence is determined by the real-time power conditions and the transfer mechanism's design. Of note, when an outage of the normal source occurs with a Closed Transition Transfer Switch, transfer from the normal to emergency sources will be an open transition sequence with a momentary power interruption, necessarily so because only the emergency source is live. Later, retransfer to normal will occur without interruption because both power sources are live.

Importantly, an ATS needs to complete transfers even under fault conditions because fault currents must be cleared by overcurrent protection devices. For this reason, the testing prescribed by listing agencies verifies close-on ratings for automatic transfer switches.

Function 5: Sense Restoration of the Normal Source

While connected to the emergency source, the ATS monitors the condition of the normal source. When acceptable power returns, the transfer switch initiates retransfer from the emergency source to the normal source.

Function 6: Retransfer Load to the Normal Source

After sensing restoration of acceptable power, an ATS initiates retransfer to the normal source. In this instance, both power sources are available and acceptable. Typically, retransfer occurs when the voltages, frequencies, and (where applicable) phase angles of the two sources are within 5 percent, 2 Hertz, and 5 degrees, respectively. For more information, see the ASCO Power Technologies [Basic Power Source Synchronization and Paralleling video](#).



The Role of Delays

The sections above identify the essential functions of an ATS and the events that necessarily occur during every switching cycle. Not mentioned are the roles of various delay settings used in conjunction with these operations.

To differentiate whether an anomaly on a normal source is merely a short sub-cycle nuisance or a bona fide outage, ATS controllers invoke a short delay to verify that the condition is persistent. When power returns on the primary source, the ATS controller will verify that it is stable over a prescribed time delay, then retransfer to the primary source. See ASCO's [Timing Delays for ATS Transition Modes](#) for more information.

Summary

To transfer loads between power sources, automatic transfer switches must (1) carry current continuously, (2) detect power failures, (3) initiate the alternate source, (4) transfer load to the emergency source, (5) sense restoration of power to the normal source, and (6) retransfer load to the normal source. Importantly, an ATS must be able to carry any fault currents that could reach the transfer switch location so that faults can be cleared by overcurrent protection devices designed for that purpose. For more information, review the resources referenced above or contact an ASCO Power Technologies representative.

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